

Build. Code. Deliver.

Student engineering workbook

Grades 6-8 • 8 modules • 16 sessions of 45 minutes

LEGO Education Computer Science & AI Kit 6–8 (#45522, 2026)



Imaginative LEGO-style mission concept. Follow the official #45522 build resources.

Predict. Build. Test. Explain. Improve.

Design a brick-built delivery rover. Carry a paper parcel, recognize a colored delivery marker, and explain the evidence behind each improvement.

Your engineering agreement

Name: _____ Team: _____

Our goal

Carry a small paper parcel and stop in a delivery bay at a colored finish marker. Every lesson adds a tested capability to the same rover.

Work safely

Every powered trial needs teacher approval, a clear floor lane, low speed and an operator beside the stop control. Stop the program immediately on unexpected motion; use the hardware power-off procedure if software or the connection fails. The supplied recipes require a short maximum trial duration. Verify these controls on the real kit before teaching.

Keep hair and fingers clear of moving parts. Power down before changing the build. Stop on unrecognized color, disconnection or loss of control. A color marker is a delivery cue, not an obstacle detector or safety system. Follow school and manufacturer procedures.

Before every run

Clear floor lane. Secure cargo. Sensor unobstructed. Known program. Driver and observer ready. Teacher approval. Stop and power controls accessible.

Keep useful evidence

Record unsuccessful trials too. Change one factor at a time. Keep the start position, floor, load and program fixed within a trial set.

Before we begin

One robot I know, its job, and an input and output it uses:

Our team stop procedure and roles:

Wake up your rover

Build a stationary input-to-output response

Your target

Identify input, rule and output, and demonstrate three predictable color-to-screen responses without wheel motion.

Gather

Kit #45522, charged double motor, color sensor, connection card, device, red and white sample cards. Wheels raised for setup.



Mission concept art; not an exact kit assembly or parts list.

Understand

An input provides evidence. A program applies a rule. An output changes a light, display or motor. Today the sensor reports a color while all motors remain stopped.

Build and test

1. Use the teacher-checked wheeled base. Label the double motor, color sensor and connected device; draw the flow of information.
2. Connect the team hardware in Coding Canvas and keep all movement blocks out of this first program.
3. Build the lesson 1 recipe: read color and show READY for white, DELIVERY for red, and CHECK for any other reading.
4. Present red, white and an unrecognized sample while stationary. Record the expected and observed messages. All wheels must stay still.
5. Change one displayed message to suit your team. Explain why changing a message does not change the sensor.

Success criteria

Three correctly explained input-to-screen tests with motors stationary.

Use block recipe: `code/lesson-1.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change only the sample color; keep lighting and sensor position fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Sample / trial	Predicted message	Observed message	Wheels stopped?
1			
2			
3			

Explain and improve

A remotely operated exploration robot is still a robot. What extra capability would make this rover autonomous?

My evidence supports this change / next test:

Motor checkout

Make a controlled, repeatable drive

Your target

Run a bounded forward drive, demonstrate stop, and identify two sources of variation.

Gather

Teacher-checked LEGO wheeled base, paired device, ruler and taped floor lane with clear run-out space.



Mission concept art; not an exact kit assembly or parts list.

Understand

Each drive wheel has a motor. Two similar wheel speeds move the rover approximately straight; unequal speeds create a turn. A speed setting is a motor command, not a measurement in centimetres per second.

Build and test

1. With power off, check wheel clearance and identify the teacher-demonstrated stop and power controls.
2. Using the lesson 2 recipe, set a low movement command, such as 20 percent, for 0.8 seconds, followed by stop. Pretest with wheels raised.
3. Mark a start line, place the rover on the floor and start from the device. Measure travel after the automatic stop. Repeat three times with the same settings.
4. With teacher supervision, test the on-screen stop during a short drive. Verify the hardware power-off fallback before student trials.
5. Check surface, wheel alignment and charge if the path curves. Change one setting and retest.

Success criteria

Three bounded drives with automatic stops, plus an operator-stop test and distance records.

Use block recipe: `code/lesson-2.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change only one factor, such as commanded speed. Keep time, start direction, floor and load fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / speed setting	Time (ms)	Distance (cm)	Drift / stop notes
1			
2			
3			

Explain and improve

Why can the same motor command produce a different distance tomorrow?

My evidence supports this change / next test:

Design a cargo cradle

Carry a paper parcel without blocking the robot

Your target

Build and compare two removable cargo cradles using consistent trials and a stated design constraint.

Gather

Teacher-checked LEGO rover, paper parcel, light cardboard, removable adhesive and spare kit bricks approved for the attachment.



Mission concept art; not an exact kit assembly or parts list.

Understand

A chassis supports components. An attachment changes how weight is distributed and may obstruct sensors or wheels. Engineers compare designs against constraints, rather than assuming that a stronger-looking structure performs better.

Build and test

1. Make the same lightweight paper parcel for both designs. Sketch a low cradle that leaves the color sensor, wheels and stop controls clear.
2. Build a removable cradle on the available upper mounting area, following manufacturer restrictions. Do not tape across circuitry, wiring or sensor faces. Teacher checks the fit with power off.
3. Use lesson 3's short drive. Test the same distance/time setting three times. Record whether the parcel stays in place and whether the attachment rubs or tips.
4. Revise one feature, such as the height of a paper retaining wall or cradle position. Repeat on the same floor with the same parcel and code.
5. Choose a design using your results. Draw its attachment points so another team can reproduce it.

Success criteria

Parcel stays in place for three short trials; the sensor view, wheels, controls and switch remain unobstructed.

Use block recipe: `code/lesson-3.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change one cradle feature. Keep parcel, surface, start position and program fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Design / trial	Parcel retained?	Any rubbing or tipping?	Evidence / revision
1			
2			
3			

Explain and improve

Why should you keep the parcel and code unchanged when comparing the cradles?

My evidence supports this change / next test:

Crack the color code

Build and test a dependable marker reader

Your target

Compare repeated color classifications and select a marker setup using recorded evidence.

Gather

Stationary LEGO rover with downward-facing color sensor, matte red and white samples, one unfamiliar sample, ruler and log.



Mission concept art; not an exact kit assembly or parts list.

Understand

A color sensor classifies reflected light. The reported color can change with lighting, height, surface and angle. Test the actual marker materials before using readings to control movement.

Build and test

1. Keep motors stopped. Use the teacher-approved sensor mount and measure its height above the sample.
2. Open the lesson 4 recipe to display the reported color repeatedly. Make a prediction for each sample.
3. Present red, white and an unfamiliar surface three times each at the same height. Record every reported color, including no-color or unknown.
4. Change only height OR lighting, then repeat the same samples. Identify misclassifications; do not discard them.
5. Choose a red finish strip and white approach surface that the sensor distinguishes reliably in these tests. Repeat on the actual lane before powered use.

Success criteria

Nine initial readings, a controlled comparison, and a justified marker choice with a limitation.

Use block recipe: `code/lesson-4.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change sensor height OR lighting; hold the sample, angle and other conditions fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Sample / condition	Reading 1	Reading 2	Reading 3
White			
Red			
Other			

Explain and improve

The same red card is reported correctly only twice in three trials. What should you do before driving?

My evidence supports this change / next test:

Code a delivery route

Drive to a marked delivery bay

Your target

Program a two-leg route and use three trials to tune one timing variable.

Gather

LEGO rover with empty cargo cradle, device, floor tape and ruler. Mark a wide delivery bay and a clear run-out area.



Mission concept art; not an exact kit assembly or parts list.

Understand

A sequence runs commands in order. Timed movement is open-loop: the program does not check where the rover actually ended up. A turn duration is not a guaranteed angle. Calibrate movement on the same floor used for the mission.

Build and test

1. In the online Mission Lab, predict a route before running it. Try forward, turn and stop commands. The grid is a logic model, not a physical calibration tool.
2. On the floor, mark a start, a turn point and a delivery bay. Sketch the intended route with arrow directions.
3. Use the lesson 5 block recipe. Start with short forward times and a low-speed timed spin; test the turn alone before combining the route. A 0.4 second spin is only an initial tuning value.
4. Run the two-leg route three times. Measure how far the stopping point is from the bay centre. Change only turn time or one drive time, then retest.
5. Place a repeat block around a short segment only if your chosen route repeats. Explain why blindly repeating an inaccurate segment compounds error.

Success criteria

A documented two-leg route ends within the team's stated bay tolerance in at least two of three trials, and stops every time.

Use block recipe: `code/lesson-5.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change one timing value. Keep speeds, floor, start alignment and load fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / timing change	End error (cm)	Turn observation	Inside bay?
1			
2			
3			

Explain and improve

Why can a route succeed once but miss the bay on the next run?

My evidence supports this change / next test:

Nail the delivery stop

Stop on a colored finish marker

Your target

Use a repeated color decision to stop on a red marker in three supervised trials and measure stopping position.

Gather

LEGO rover with checked downward-facing color sensor, matte white lane, broad red finish strip, ruler and empty run-out area.



Mission concept art; not an exact kit assembly or parts list.

Understand

A conditional chooses an action from a reading. While white is detected, take one brief, low-speed movement step; red or any unrecognized reading ends the trial. Keep the stop latched: only a new operator start can begin again.

Build and test

1. Use the stationary tests from mission 4 to approve the white lane and red strip. Place the strip well before the end of the clear lane.
2. Build the lesson 6 recipe: stop first; repeat at most 20 times; read color; if white, move slowly for 0.1 seconds then stop; otherwise end the trial. End with stop.
3. Test the rule while stationary for white, red and unknown samples before enabling motion. Red and unknown must never command movement.
4. Run on the floor with the teacher beside the controls. Measure where the sensor stops relative to the leading edge of the red strip. Record overshoot or a missed marker as a failed trial.
5. Repeat three times. If detection is late, lower speed or widen the marker, changing one factor per comparison. Keep the stop latched and the bounded repeat.

Success criteria

Three marker-triggered stops with measured positions, no automatic restart and no motion on an unrecognized reading.

Use block recipe: `code/lesson-6.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Change speed OR marker width; keep sensor height, light, start and surface fixed.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / setting	Marker recognized?	Overshoot (cm)	Stopped / failure
1			
2			
3			

Explain and improve

Why can a narrow red strip be missed even when the stationary test worked?

My evidence supports this change / next test:

Plan the delivery mission

Deliver a paper parcel using a tested approach

Your target

Combine a cargo attachment, a bounded route and a sensor stop into a mission with measurable acceptance criteria.

Gather

LEGO rover, tested cargo cradle and paper parcel, white approach lane, red marker in a delivery bay, ruler and worksheet.



Mission concept art; not an exact kit assembly or parts list.

Understand

A mission combines requirements, constraints and evidence. Reliable behaviour is more valuable than speed. A state is a named stage such as ready, approaching or stopped; only one stage should control the motors at a time.

Build and test

1. Write three criteria before testing: parcel retained, stopped inside the bay, and no contact. Use your measured stopping positions to locate the red strip.
2. Rotate four roles. Sketch the lane, mark the start and identify the immediate stop controls. Teacher approves the setup.
3. Use the bounded color-stop recipe from lesson 6 as the final approach. Add a short calibrated route segment only after testing it separately.
4. Run three identical trials. A trial passes only when all three criteria hold. Record failed attempts too.
5. Change one factor based on evidence and retest three times. Keep separate records for each version.

Success criteria

At least two of three full mission trials meet all three criteria; every trial remains bounded and supervised.

Use block recipe: `code/lesson-7.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Choose one deliberate change between trial sets, then keep it fixed within each set.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Trial / version	Parcel retained?	In bay / no contact?	Pass? Evidence
1			
2			
3			

Explain and improve

Your first version passed 1 of 3 trials and your revision passed 2 of 3. What can you claim?

My evidence supports this change / next test:

Showcase with evidence

Prove, explain and hand over your robot

Your target

Present a reproducible demonstration and defend a design choice using measurements and a limitation.

Gather

Final LEGO rover, unchanged marked lane, saved Coding Canvas project, log and handover sheet.



Mission concept art; not an exact kit assembly or parts list.

Understand

An engineering handover lets another person understand, operate and evaluate a design. Clear documentation includes the actual limits and remaining problems. A demonstration shows one run; a test log shows what happened across runs.

Build and test

1. Freeze a named program version and draw the final layout. Write setup, start, stop and reset instructions for another team.
2. Run three final acceptance trials with the same parcel and target. Record all results before selecting a demonstration clip or photo.
3. Have another team follow your instructions under teacher supervision. Observe unclear steps and revise the handover without changing the test data.
4. Give a two-minute presentation: need and criteria; sense–decide–act explanation; one design change with before/after evidence; one limitation and next test.
5. Complete a peer review and individual reflection. Save the Coding Canvas project and export the worksheet or log. Teacher assesses each student’s explanation as well as team performance.

Success criteria

A reproducible handover, three recorded trials, a two-minute evidence-based explanation and an individual reflection.

Use block recipe: `code/lesson-8.md`. Assemble and pretest in Coding Canvas.

Predict, test, improve

Name: _____ Date: _____ Version: _____

Predict

I think the rover will... because...

Plan a fair test

Keep the final design fixed during acceptance tests. Label any later change as a new version.

One thing I will change: _____

Sketch the setup, attachment or route

Record your evidence

Final trial	All criteria passed?	Observed failure / gap	Next test
1			
2			
3			

Explain and improve

What should you say if the showcase demonstration fails?

My evidence supports this change / next test:

Final mission handover

Program version / saved project: _____

Our mission criteria and layout

Setup → start → stop → reset

Two-minute showcase

0:00-0:20: need and criteria. 0:20-0:50: input, decision, output. 0:50-1:30: before/after evidence. 1:30-2:00: limitation and next test.

Peer review

One instruction I could follow: _____

One instruction that needs clarification: _____

Individual reflection

A choice I made, the evidence I used, and what I would test next:

Budget your robotics lab

Current kit: LEGO Education Computer Science & AI 6–8 (#45522). US dollars; asking prices and planning allowances are separated below.

Purchase / allowance	USD	Basis
One kit · up to 4 learners	\$529.95	Manufacturer asking price
Six individual kits · 24 learners	\$3,179.70	6 × \$529.95; fewer charging extras than bundle
Classroom bundle · 24 learners	\$3,499.00	6 kits plus charging and replacement extras
Pilot supplies allowance	\$35.00	Planning estimate: paper, tape, ruler and cargo materials
Class supplies allowance	\$120.00	Planning estimate for six teams
Pilot total before tax / shipping	\$564.95	1 kit + \$35 supplies; existing device assumed
Class total before tax / shipping	\$3,619.00	1 bundle + \$120 supplies; 6 existing devices assumed

Prices were checked on the manufacturer's US page. Recheck before ordering. Tax, delivery, devices, school IT support and teacher training are excluded. The supplied course and browser simulator require no paid course subscription. No purchase has been made.

What the classroom bundle adds

The 24-student bundle includes six kits, three additional multi-chargers, eight extra connection cards and a replacement brick pack. Six single kits cost \$319.30 less, but do not provide the same bundle extras.

Choose a purchase plan

Pilot first: one kit, one existing device and four rotating roles. A 24-student class uses six teams of four. For pairs, plan 12 kits: \$6,359.40 for kits alone. An optional spare kit adds \$529.95. All of these are one-time hardware planning figures, not per-student subscription fees.

Older kits are a separate pathway

EV3 and Robot Inventor use different hardware and software. Do not buy them expecting these #45522 recipes to run unchanged. SPIKE direct sales ended June 30, 2026; LEGO lists app support through June 30, 2031. Check used-kit completeness, charging equipment and software before any legacy purchase.

[Selected kit and current US pricing](#)

[SPIKE retirement and support](#)